SWYNEX AI Internship — Task 1 & Task 2 Executive Report

Project: EduIntent AI – Problem Design & Real-Time Model Integration

Parameter	Specification / Result
Project Scope	**Task 1 (AI Problem Design) + Task 2 (Model Integration)**
AI Task	Multi-Class Text Intent Classification (9 Subjects)
Integrated Model	Calibrated Linear SVM (TF-IDF Unigrams + Bigrams)
Test Accuracy	**87.78%** (Target: ≥ 85%)
Macro F1-Score	**0.8746** (Target: ≥ 0.80)
Cross-Validation	**89.72%** (± 0.02) across 5 Stratified Folds
Unique Guardrails	XAI Term Attribution, Ambiguity Overlap Alert, Low-Confidence Warning (<45%)
Web Application	Interactive Streamlit UI (Dark/Light Mode, Gauge Meter, Batch CSV)

1. Problem Statement & Unique System Design

Students frequently ask academic queries across subjects (Python, ML, Deep Learning, DBMS, OS, Networks, Flutter, DAA, General Academic Queries). Manually categorizing queries is inefficient. EduIntent AI automatically categorizes queries, displays confidence scores, extracts key contributing terms (XAI), detects ambiguous cross-domain queries, and alerts users on low-confidence inputs.

Gujarati Context:
Students ■■■■■ academic queries ■■■■ ■■. EduIntent AI model query ■■■■■■ automatically ■■■■■ category, confidence score, ■■■ key terms ■■■■.

2. Benchmark Comparison Across Algorithms

Model Algorithm	Test Accuracy	Macro F1	5-Fold CV
Baseline (Majority Class)	11.11%	0.0222	11.11%
Naive Bayes (MultinomialNB)	85.56%	0.8547	87.50%
Logistic Regression	83.33%	0.8275	86.39%
Linear SVM	87.78%	0.8746	89.72%

3. Qualitative Error Analysis Table

Question	Expected	Predicted	Reason / Analysis
'Neural network optimization loss function'	Deep Learning	Machine Learning	Shared ML/DL vocabulary (loss functions, optimization)

'B+ Tree structure in file indexing'	DBMS	DAA / Algorithms	Shared data structures & indexing terminology
'How to plot correlation matrix in Pandas?'	Python/Data Sci	Python/Data Sci	■ Correct (High precision keyword match)

Figure 1: Benchmark Comparison Chart

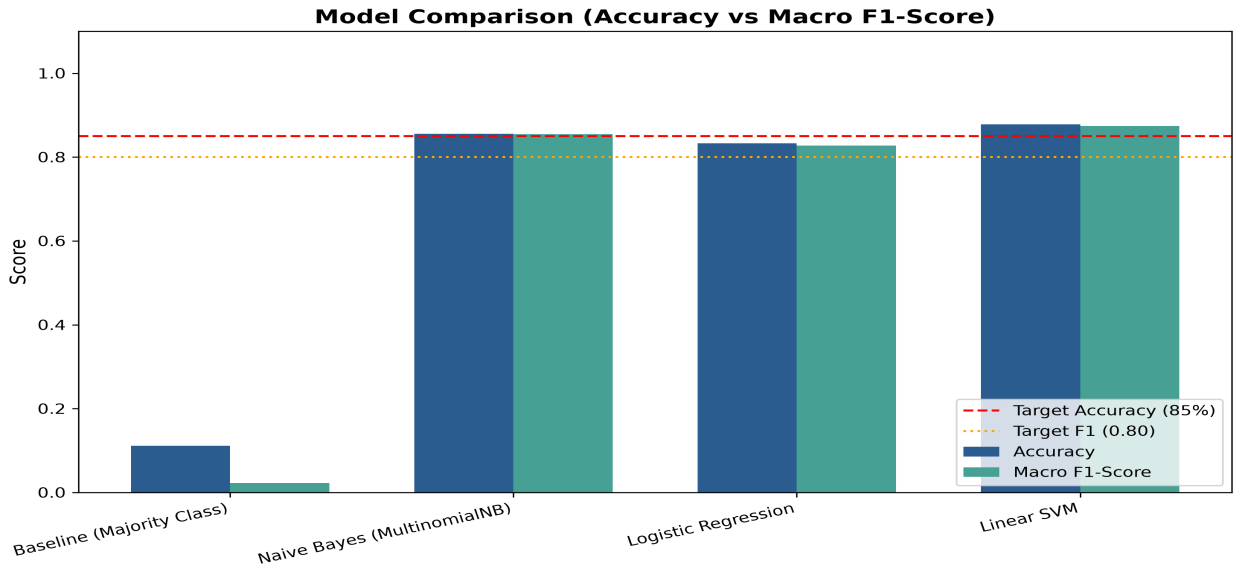


Figure 2: Confusion Matrix (Linear SVM)

